



EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

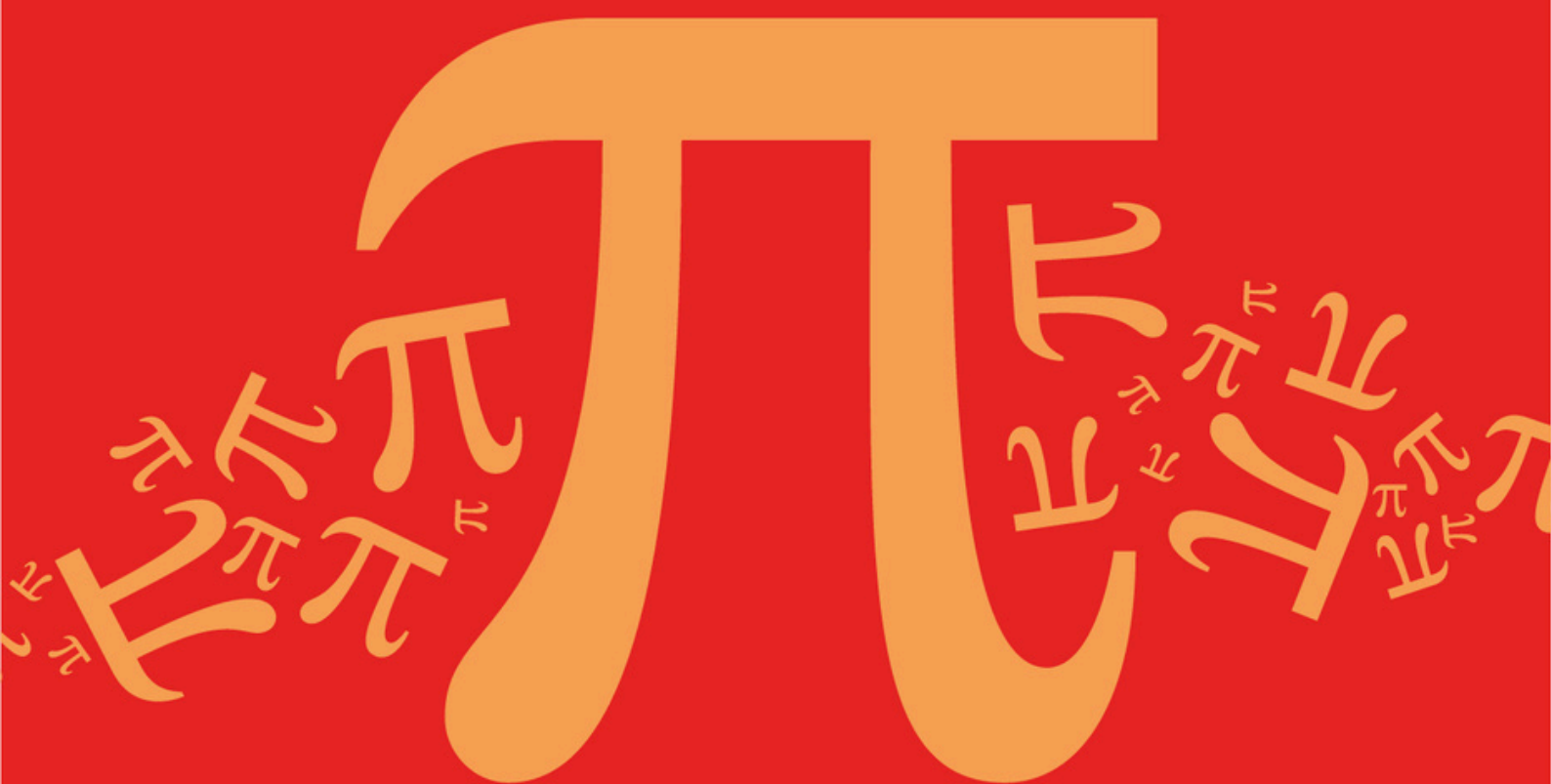
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C5: Geometric Interpretation of Determinants

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

C: Matrices

Sub topic

C5: Geometric Interpretation of Determinants

EXAM PAPERS PRACTICE

© 2024 Exams Papers Practice. All Rights Reserved

**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

It is given that **A** and **B** are 3×3 matrices such that

$$\det(\mathbf{AB}) = 24 \text{ and } \det(\mathbf{A}^{-1}) = -3$$

- (a) State the value of $\det \mathbf{A}$.

(1)

- (b) A three-dimensional shape S , with volume 20 cm^3 , is transformed using matrix **B**.

Find the volume of the image of S .

(3)

(Total 4 marks)

Q2.

The plane transformation T is defined by

$$T : \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- (a) A shape has an area of 3 square units. Find the area of the shape after being transformed by T .

(2)

- (b) (i) Find the equations of all the invariant lines of T .

(5)

- (ii) State the equation of the line of invariant points of T .

(1)

(Total 8 marks)

Q3.

The plane transformation T is represented by the matrix $\mathbf{M} = \begin{bmatrix} -3 & 8 \\ -1 & 3 \end{bmatrix}$.

- (a) The quadrilateral $ABCD$ has image $A'B'C'D'$ under T .

Evaluate $\det \mathbf{M}$ and describe the geometrical significance of both its sign and its magnitude in relation to $ABCD$ and $A'B'C'D'$.

(3)

- (b) The line $y = px$ is a line of invariant points of T , and the line $y = qx$ is an invariant line of T .

Show that $p = \frac{1}{2}$ and determine the value of q .

(5)

- (c) Find the 2×2 matrix **R** which represents a reflection in the line $y = \frac{1}{2}x$.

(2)

(Total 10 marks)

Q4.

The plane transformations T_A and T_B are represented by the matrices **A** and **B**

respectively, where $\mathbf{A} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 2 & 5 \\ 5 & 13 \end{bmatrix}$.

- (a) Find the equation of the line which is the image of $y = 2x + 1$ under T_A .

(3)

- (b) The rectangle $PQRS$, with area 4.5 cm^2 , is mapped onto the parallelogram $P'Q'R'S'$ under T_B . Determine the area of $P'Q'R'S'$.

(2)

- (c) The transformation T_C is the composition

' T_B followed by T_A '

Find the matrix which represents T_C , give a full geometrical description of T_C .

(5)

(Total 10 marks)

Q5.

The plane transformation T is defined by

$$T: \begin{bmatrix} x' \\ y' \end{bmatrix} = \mathbf{M} \begin{bmatrix} x \\ y \end{bmatrix}$$

where $\mathbf{M} = \begin{bmatrix} -1 & 4 \\ -1 & 3 \end{bmatrix}$.

- (a) Evaluate $\det \mathbf{M}$ and state the significance of this answer in relation to T .

(2)

- (b) Find the single eigenvalue of $\mathbf{M}$ and a corresponding eigenvector. Describe the geometrical significance of these answers in relation to T .

(5)

- (c) Show that the image of the line $y = \frac{1}{2}x + k$ under T is $y' = \frac{1}{2}x' + k$.

(3)

(Total 10 marks)

Q6.

A transformation T of three-dimensional space is given by the matrix $\mathbf{W} = \begin{bmatrix} 3 & -1 & 1 \\ 2 & 0 & 2 \\ -1 & 1 & 1 \end{bmatrix}$.

- (a) (i) Evaluate $\det \mathbf{W}$, and describe the geometrical significance of the answer in relation to T .

(2)

- (ii) Determine the eigenvalues of $\mathbf{W}$.

(6)

- (b) The plane H has equation $\mathbf{r} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 0$.

- (i) Write down a cartesian equation for H .

(1)

- (ii) The point P has coordinates (a, b, c) . Show that, whatever the values of a, b and c , the image of P under T lies in H .

(4)

(Total 13 marks)

Q7.

The transformation T maps (x, y) to (x', y') where

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- (a) Describe the difference between *an invariant line* and *a line of invariant points* of T .

(1)

- (b) Evaluate the determinant of the matrix $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ and describe the geometrical significance of the result in relation to T .

(2)

- (c) Show that T has a line of invariant points, and find a cartesian equation for this line.

(2)

- (d) (i) Find the image of the point $(x, -x + c)$ under T .

(2)

- (ii) Hence show that all lines of the form $y = -x + c$, where c is an arbitrary constant, are invariant lines of T .

(2)

- (e) Describe the transformation T geometrically.

(3)

(Total 12 marks)

Q8.

(a) Evaluate the determinant of the matrix $\mathbf{M} = \begin{bmatrix} 1 & 3 & -1 \\ 2 & 3 & 1 \\ 4 & 0 & -5 \end{bmatrix}$.

(2)

- (b) A three-dimensional shape S , with volume 12 cm^3 , is transformed by a transformation having matrix $\mathbf{X}$. Find the volume of the image of S in the case when:

(i) $\mathbf{X} = \mathbf{M}$;

(2)

(ii) $\mathbf{X} = \mathbf{M} \mathbf{N}^2$, where $\mathbf{N}$ is a 3×3 matrix and $\det \mathbf{N} = \frac{1}{3}$.

(2)

(Total 6 marks)

Q9.

The transformation T is a stretch in a fixed direction, and maps the point (x, y) to the image point (x', y') , where

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 6.4 & -7.2 \\ -7.2 & 10.6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- (a) Determine the scale factor of the stretch.

(2)

- (b) By considering the invariant points of T , or otherwise, find an equation for the line of invariant points of T , giving your answer in the form $y = mx$.

(5)

(Total 7 marks)

Q10.

- (a) Show that $(a + b + c)$ is a factor of

$$\begin{vmatrix} a & b & c \\ b+c & c+a & a+b \\ b-c & c-a & a-b \end{vmatrix}$$

Express this determinant as the product of $(a + b + c)$ and a quadratic factor.

(5)

- (b) Hence, or otherwise, show that there is a single real value of a for which the system of equations

$$ax + 3y + z = -5$$

$$4x + (1 + a)y + (a + 3)z = 9$$

$$2x + (1 - a)y + (a - 3)z = 15$$

does not have a unique solution, and find this value of a .

(5)

(Total 10 marks)



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.